THE CATEGORY OF MOTIVES IS KAROUBIAN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the construction of the candidate kernel from the idempotent correspondence, and the verification of the universal property using the description of morphisms.

Editorial note. The label lemma-Karoubian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-Karoubian –start 888 –end 905 –local-id 28161 returned RESULT=OK (L28161 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Description of morphisms of motives through correspondences. The construction of the category of motives, in which morphisms between motives are given by correspondences satisfying a projection condition. Identifies the projector with an idempotent correspondence and verifies the conditions defining morphisms.
- Characterization of a Karoubian category through kernels of idempotents. Homology, the definition of a Karoubian category through the existence of kernels of idempotent endomorphisms. Reduces the statement to exhibiting the kernel of the complementary projector.

Source reference. Original definition 09SG: sources/stacks-chow/tags.txt, source lines 12354–12354. Prerequisite pointer invoked at proof line(s) 904; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The category M_k is Karoubian.*

Proof. Let $M = (X, p, m)$ be a motive and let $a \in \mathrm{Mor}(M, M)$ be a projector. Then $a = a \circ a$ both in $\mathrm{Mor}(M, M)$ as well as in $\mathrm{Corr}^0(X, X)$. Set $N = (X, a, m)$. Since we have $a = p \circ a \circ a$ in $\mathrm{Corr}^0(X, X)$ we see that $a : N \to M$ is a morphism of M_k. Next, suppose that $b : (Y, q, n) \to M$ is a morphism such that $(1 - a) \circ b = 0$. Then $b = a \circ b$ as well as $b = b \circ q$. Hence b is a morphism $b : (Y, q, n) \to N$. Thus we see

Date: stacks-project:weil.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

that the projector $1 - a$ has a kernel, namely N and we find that M_k is Karoubian, see Homology, Definition 09SG. $\square$